# SIDE ALUSED

Soklite liides

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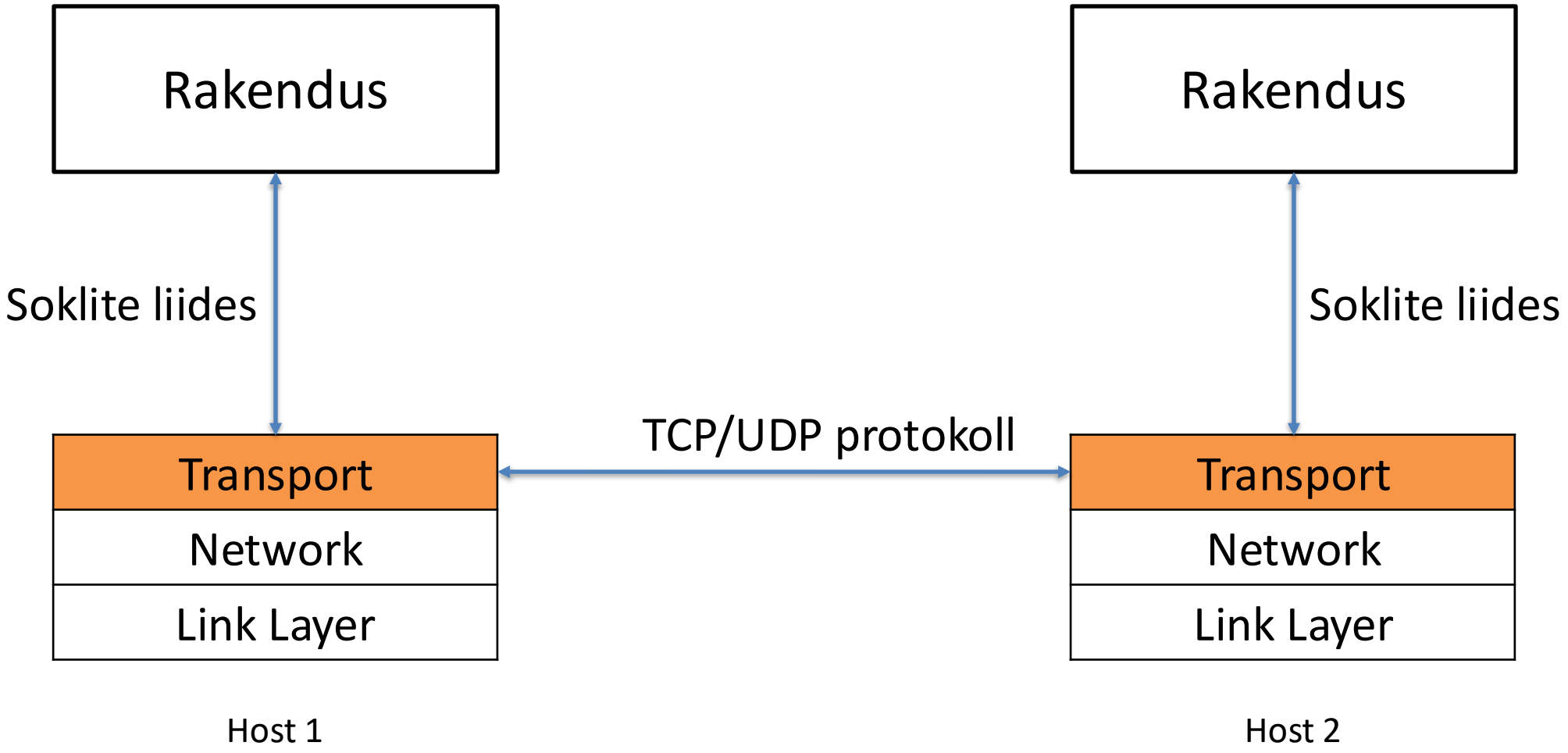
# TCP/IP transpordikihi liides



# Sokkel (*socket*)

Standardselt määratud pordinumbrid identifitseerivad üheselt rakenduse tüübi (FTP või HTTP või DNS jne), kuid neid ei saa kasutada iga sellise rakendusega seotud rakendusprotsessi üheselt identifitseerimiseks.

Rakendusprotsess määratakse kindlaks võrgu ja eraldi arvuti (IP-aadress, pordi number) paariga, mida nimetatakse **sokliks** (*socket*). IP-aadressi ja UDP-pordi numbriga määratud soklit nimetatakse **UDP-sokliks** ning IP-aadressi ja TCP-pordi numbrit **TCP-sokliks**.

# Berkeley soklid

Soklid (*socket*) ilmusid esmakordselt Berkeley UNIX 4.2 BSD (1983).

- Unixis on sokkel eriline faili liik.
- Kõik, mis faili kirjutatakse, edastatakse võrgu kaudu.
- Andmete edastamine üle võrgu on programmeerija eest varjatud.

Soklid on transpordiallsüsteemi *de facto* liidesestandard.

Erinevates operatsioonisüsteemides ja programmeerimiskeeltes on rakendatud erinevaid soklite variante.

# Berkeley soklite operatsioonid

| Operatsioon | Eesmärk |
|---|---|
| Socket | Luua uus sokkel |
| Bind | Siduda sokkel IP-aadressi ja hostiga |
| Listen | Deklareerida valmisolekut võtta vastu ühendusi |
| Accept | Võtta vastu ühenduse päring |
| Connect | Luua ühendus |
| Send | Saada andmed üle võrgu |
| Receive | Andmete saamine võrgust |
| Close | Ühenduse sulgemine |

# Klient-server mudel

Berkeley soklite koostoimed:

- Server
- Klient

Server - töötab (kuulab) teadaoleval IP-aadressil ja pordil ning ootab passiivselt ühenduse päringuid.

Klient - loob aktiivselt ühenduse serveriga määratud IP-aadressil ja pordil.

# Soklite töö

Server

Klient

# Soklite töö

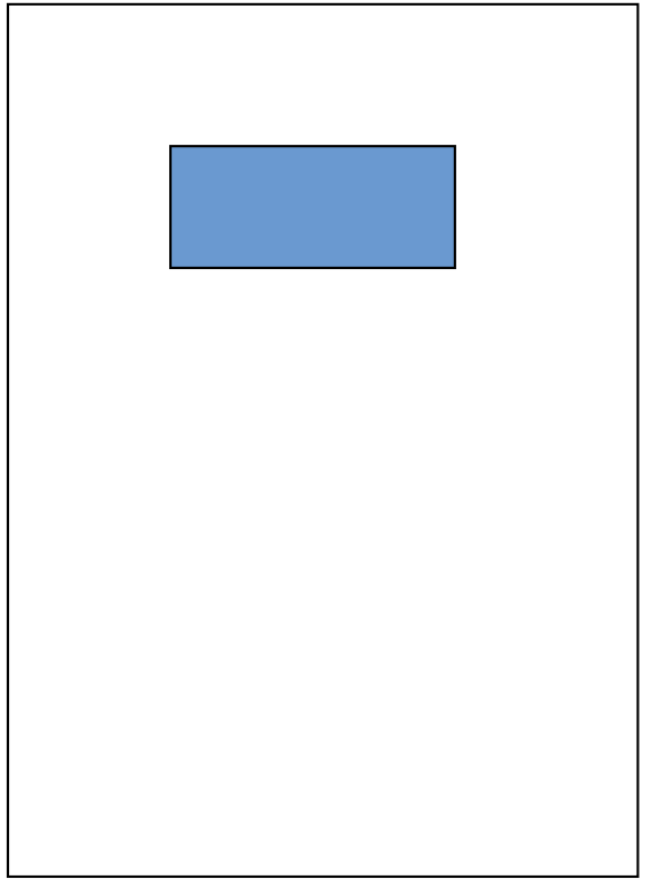
Socket

Server

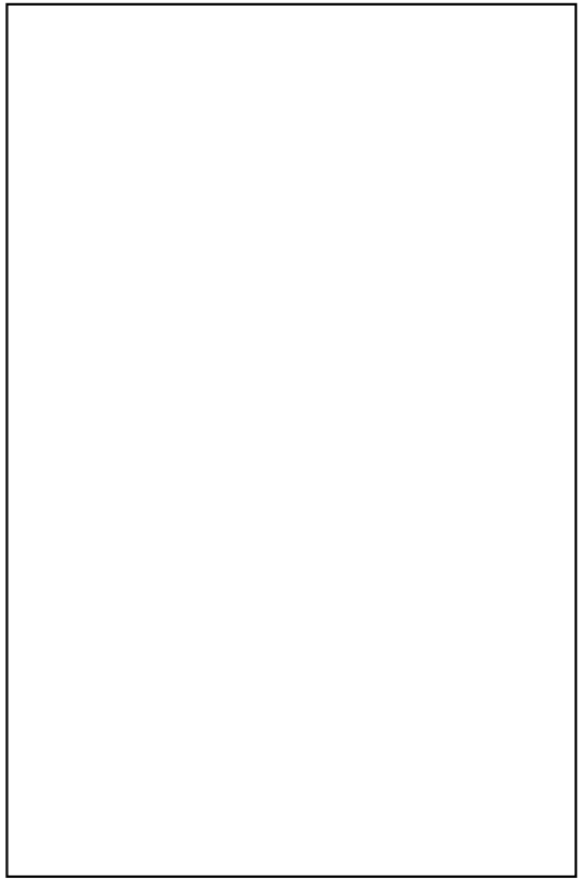
Klient

# Soklite töö

Socket



Server



Klient

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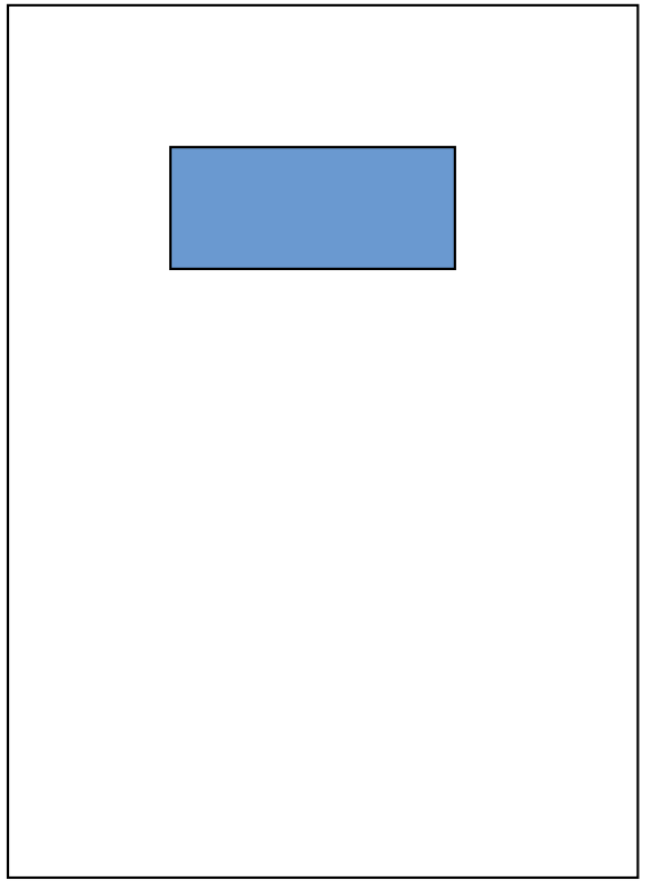
Socket

Bind



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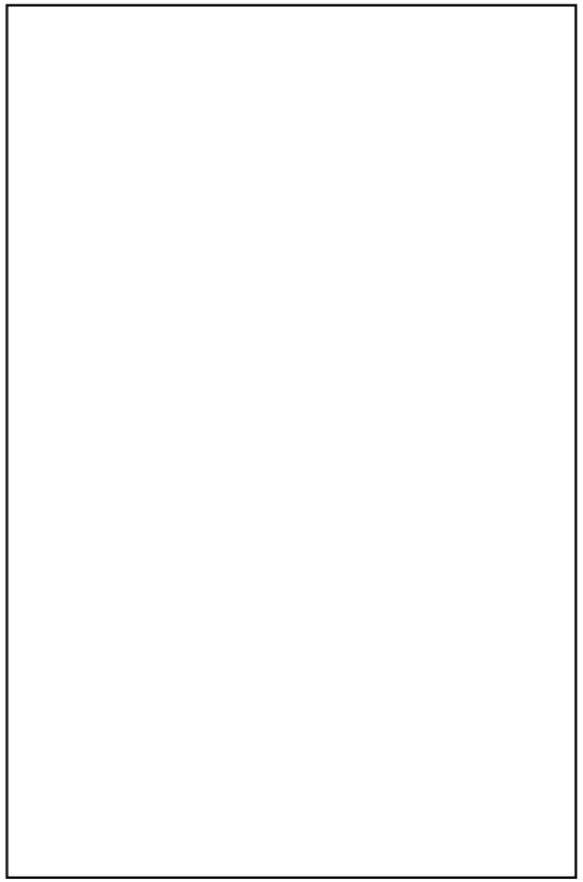

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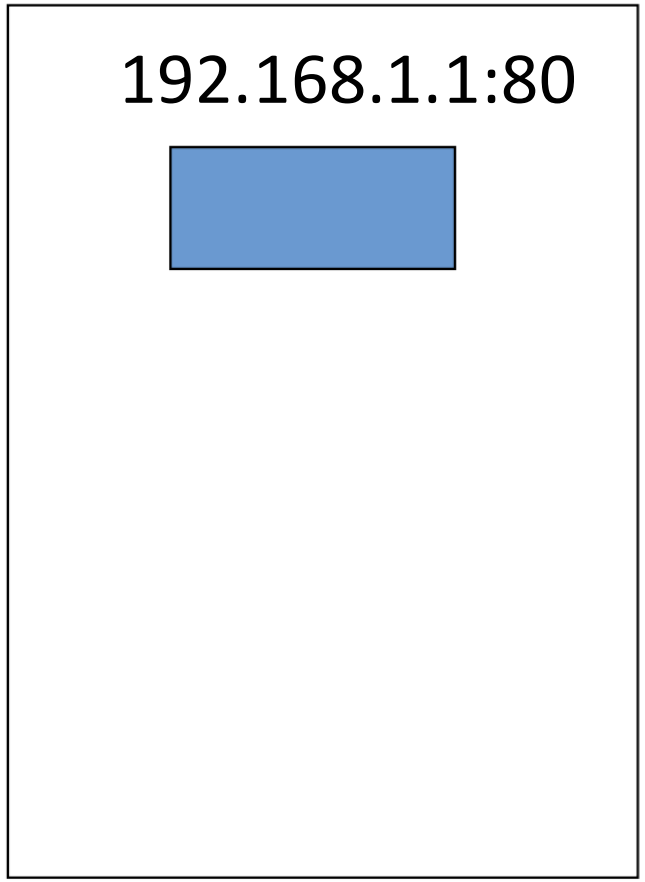


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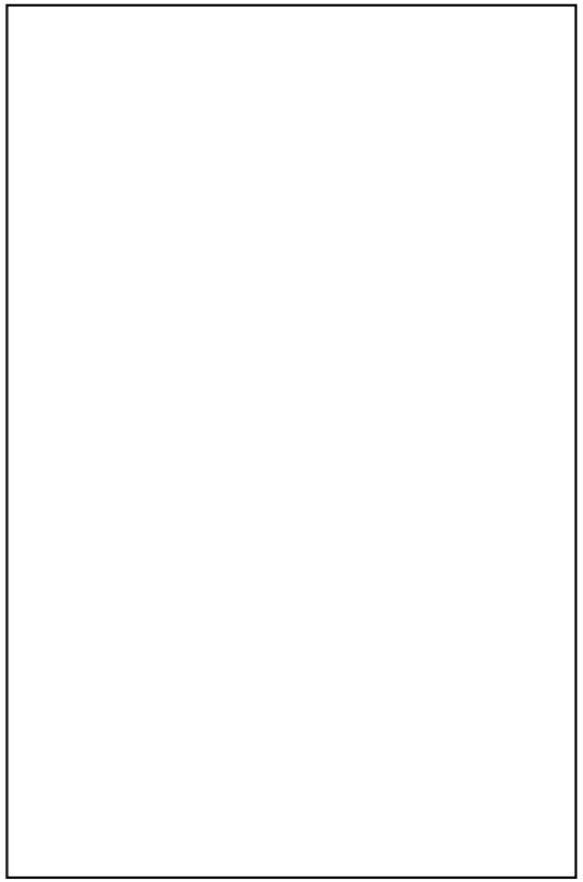

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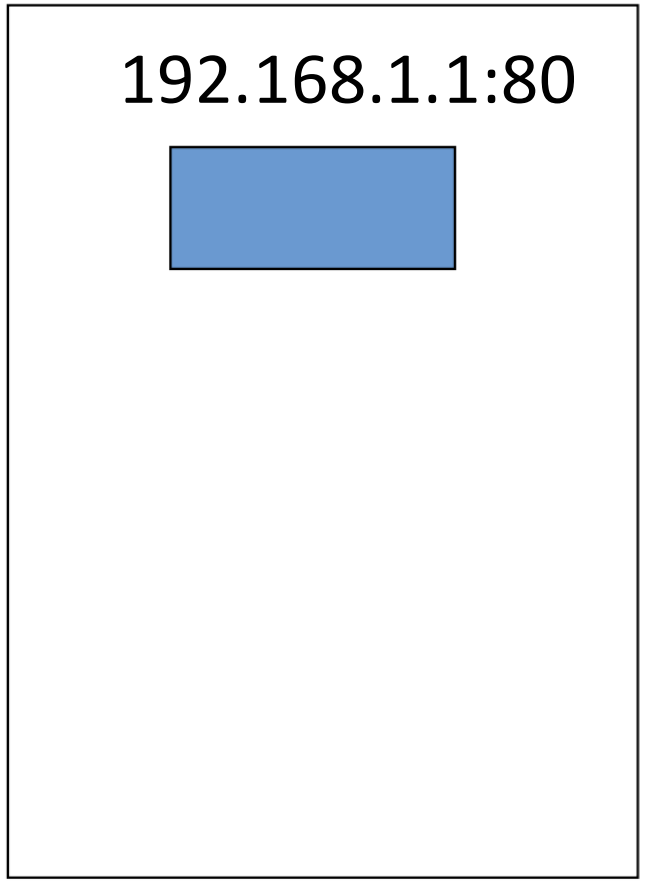


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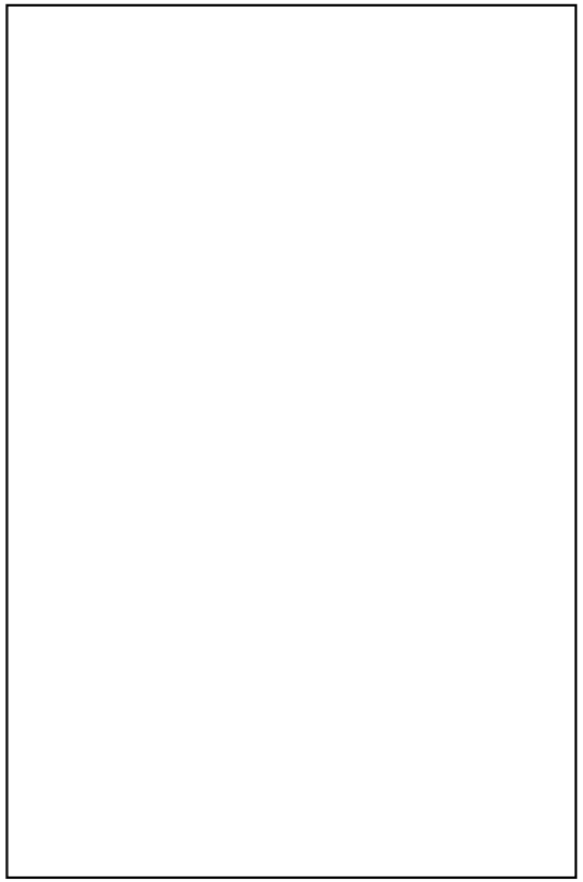

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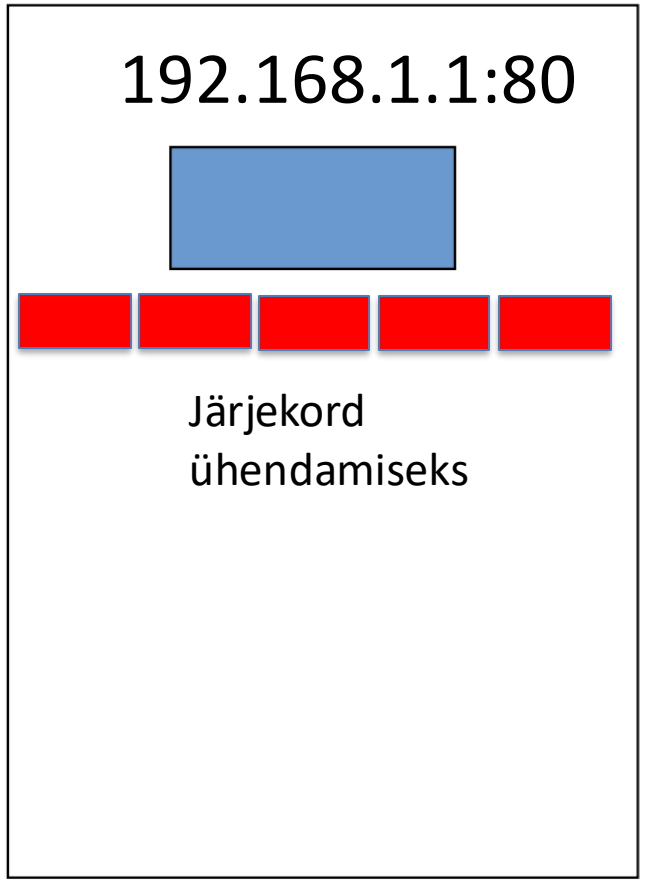


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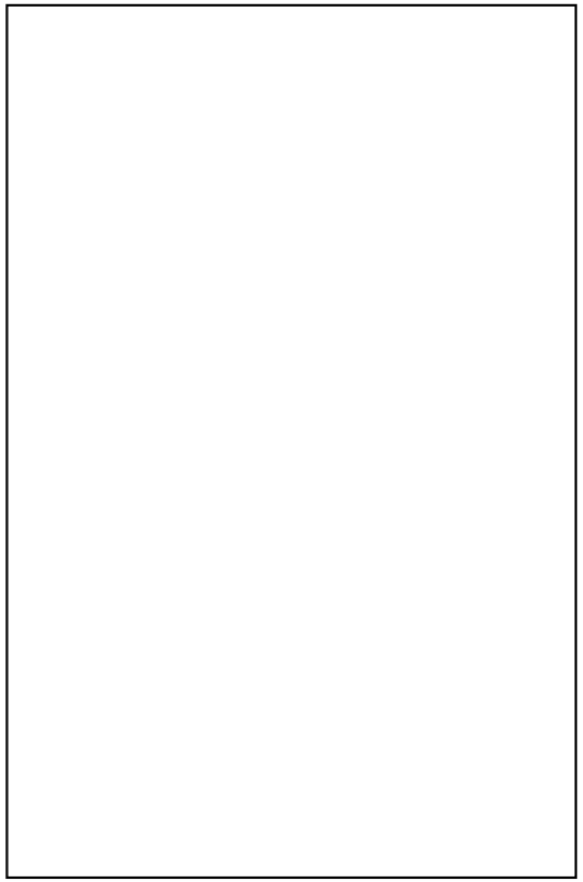

Klient

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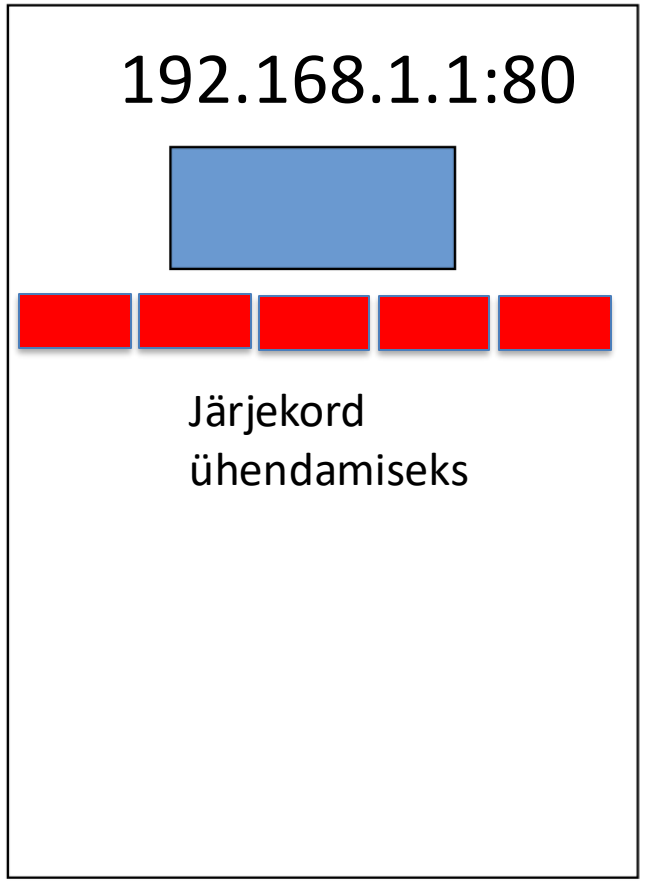


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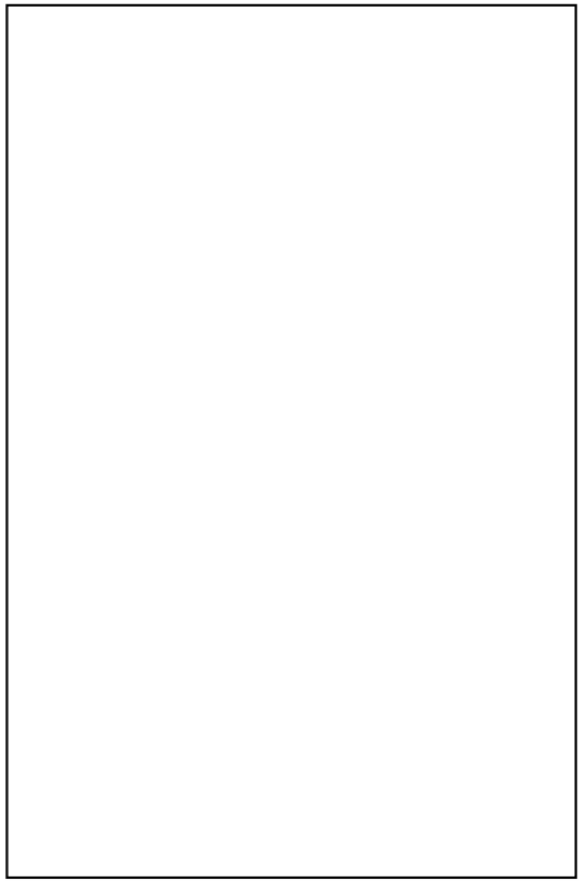

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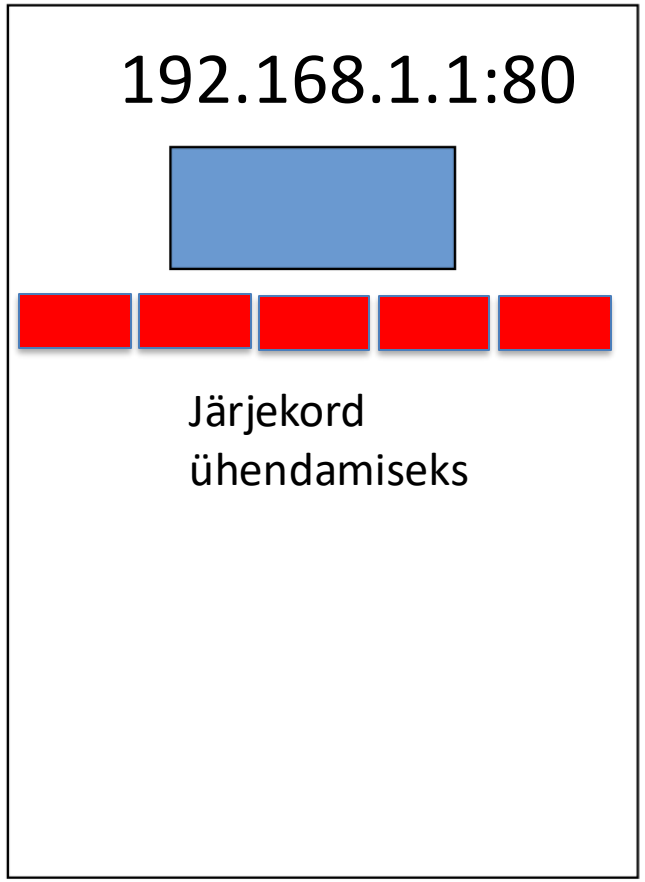


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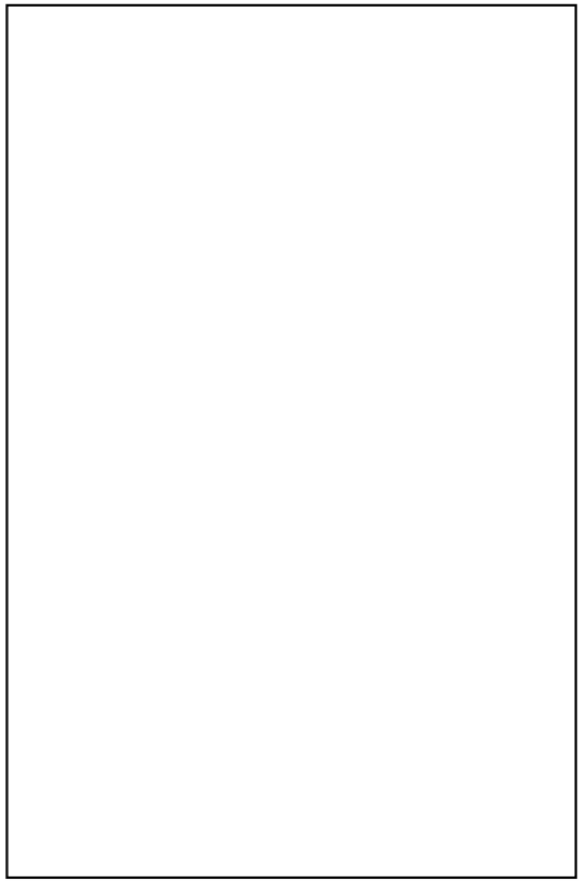

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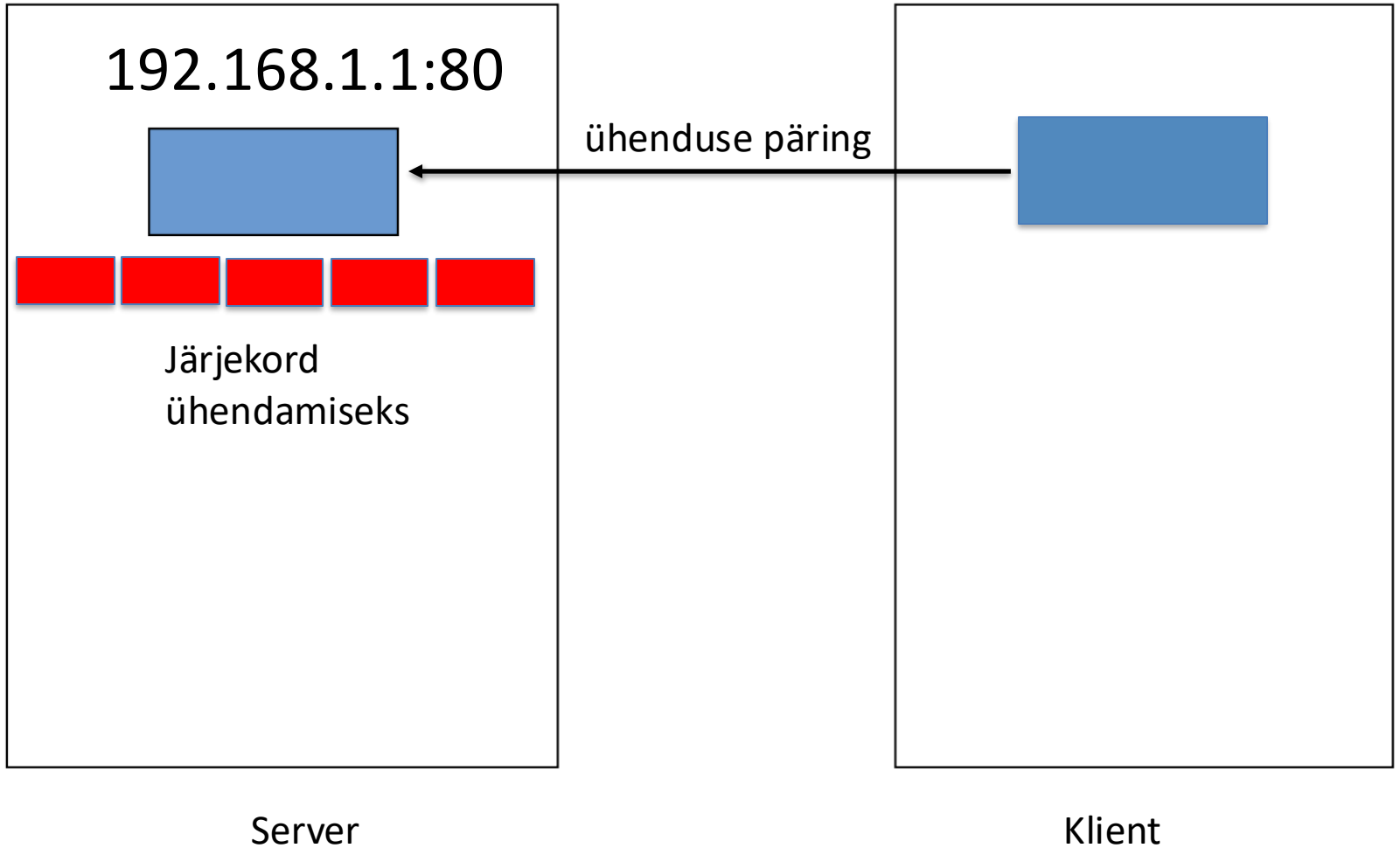


Socket

Connect

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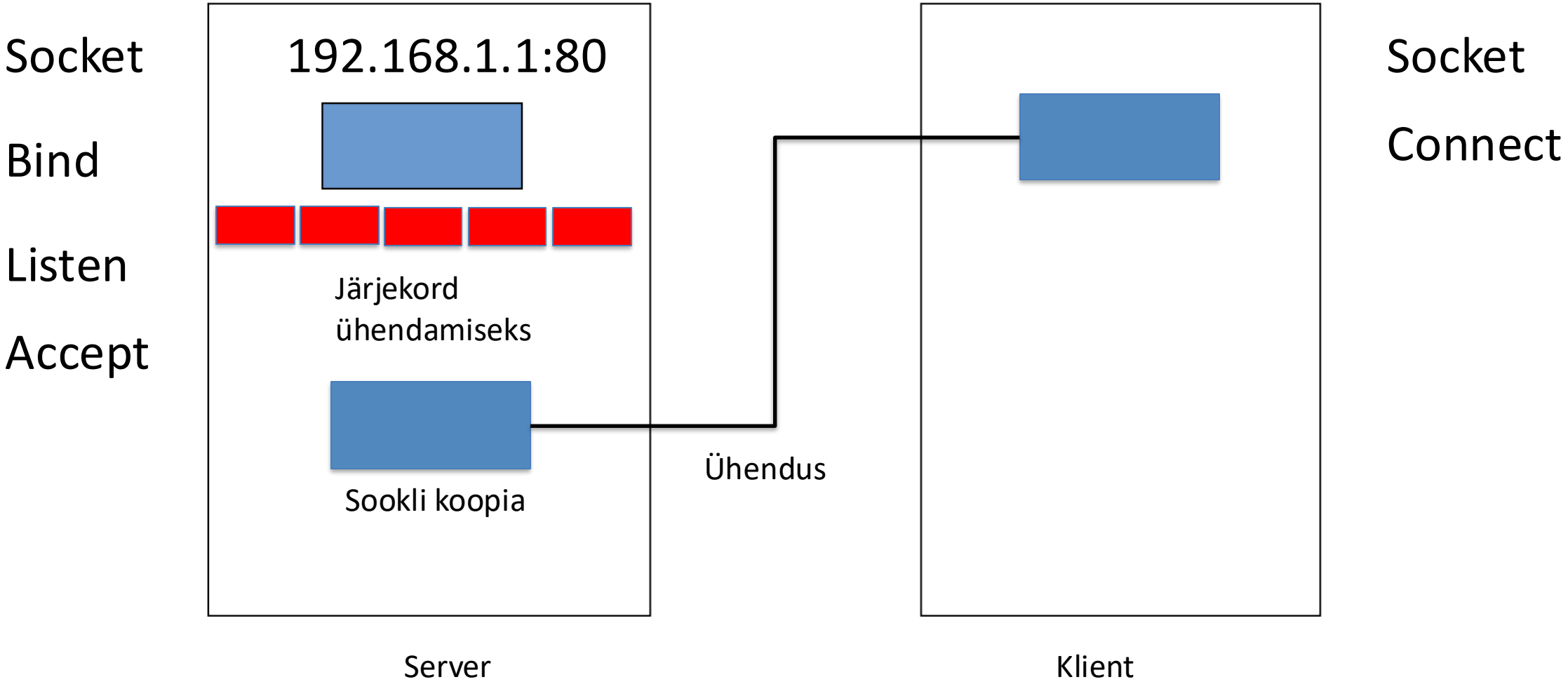
Socket
Bind
Listen
Accept
192.168.1.1:80
Järjekord ühendamiseks
Sookli koopia
Ühendus
Server
Klient
Socket
Connect

# Soklite töö

Socket

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Listen

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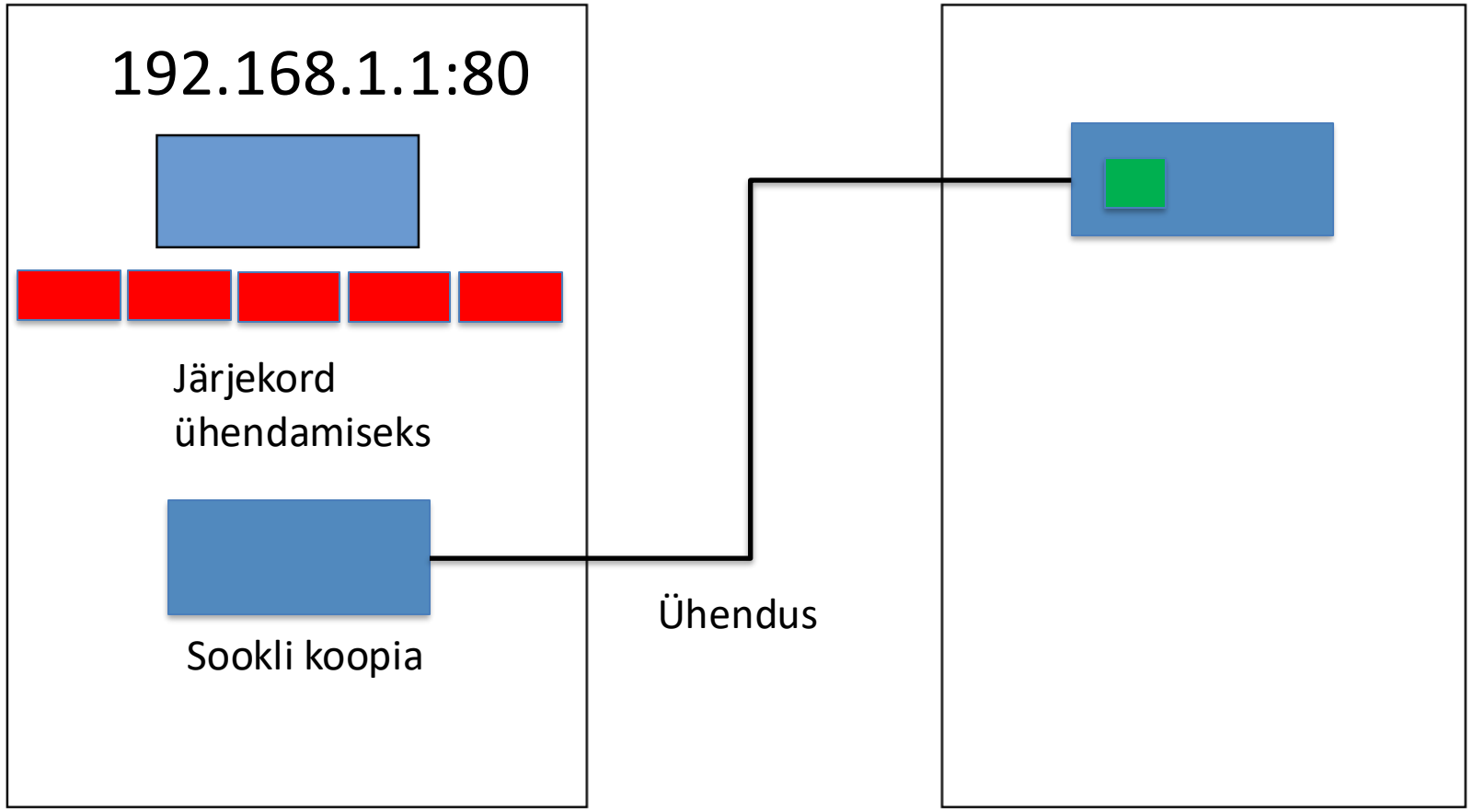


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Send

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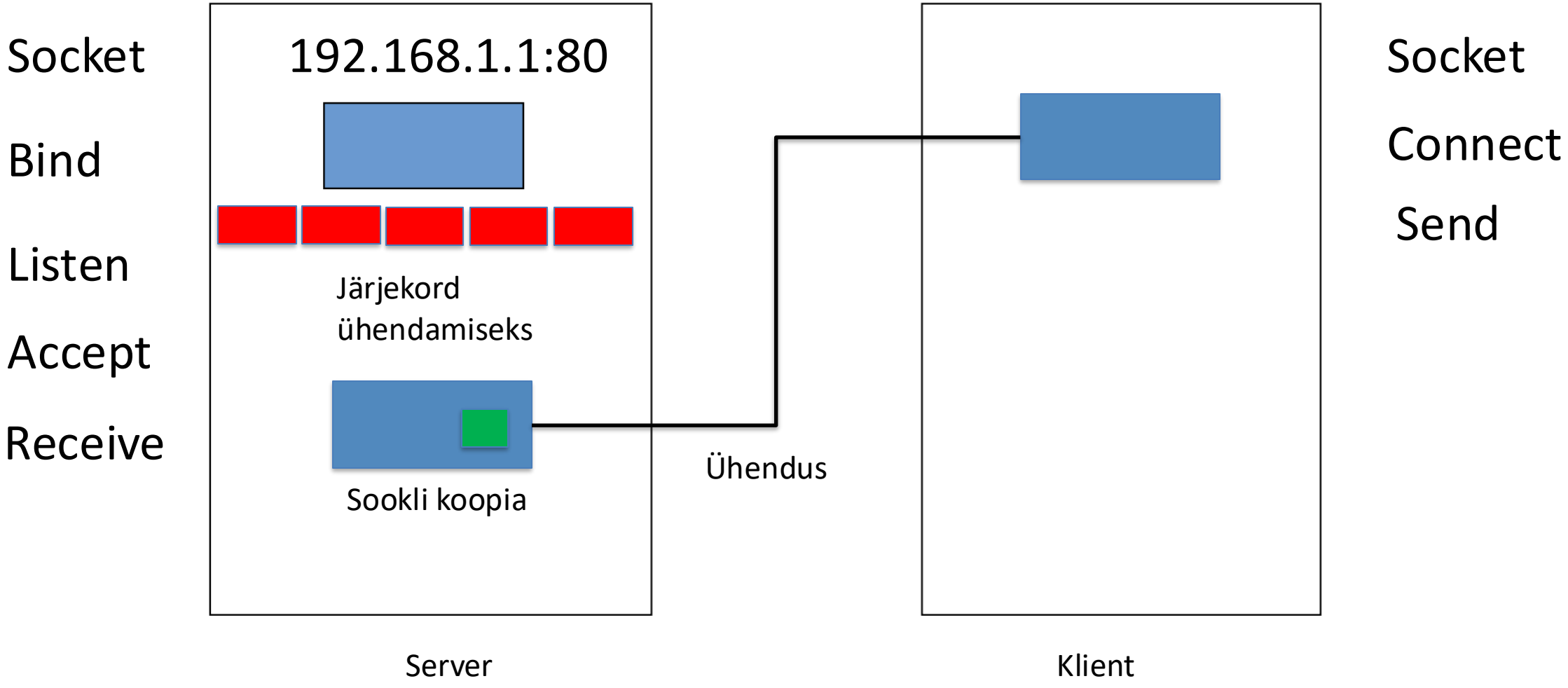
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# Soklite töö

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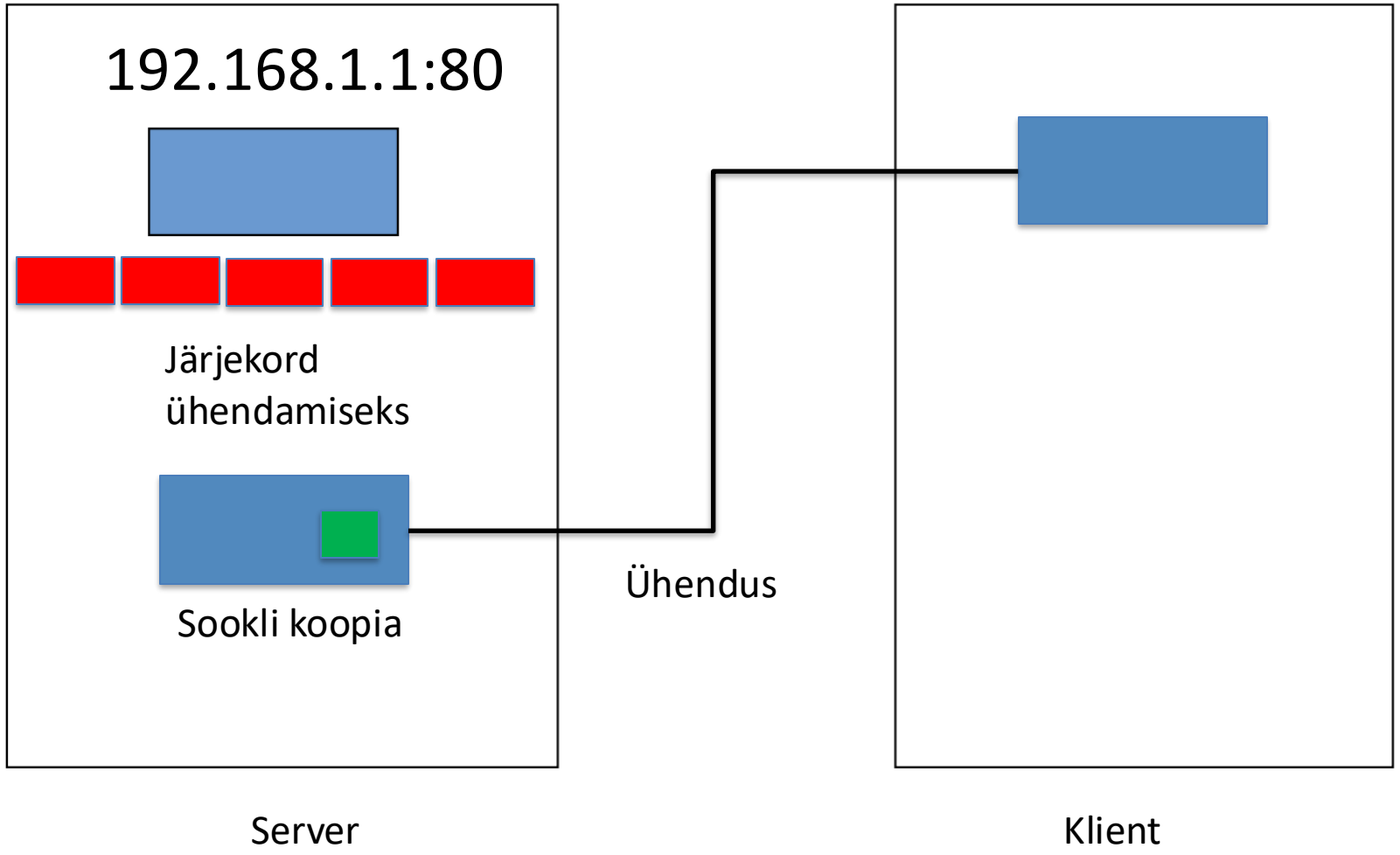


Socket

Connect

Send

Close

# Näide Pythonis Serverisokkel

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.bind(('192.168.0.1', 8888))
s.listen(1)
conn, addr = s.accept()
while True:
    data = conn.recv(1024)
    if not data: break
    conn.sendall(data)
conn.close()
```

# Soklite tüübid

Võrgukihi protokollid

- socket.AF_INET - IPV4
- socket.AF_INET6 - IPv6

Transpordikihi protokollid

- socket .SOCK_STREAM - TCP
- socket.SOCK_DGRAM - UDP

On ka teisi tüüpe, kuid neid kasutatakse harva

# Näide Pythonis Kliendisokkel

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect(('192.168.0.1', 8888))
s.sendall(b'Hello, world!')
data = s.recv(1024)
s.close()
print('Received data: ', repr(data))
```

# Kokkuvõte

Soklid on transpordikihi liidese *de facto* standard.

Programmeerija suhtleb transpordikihiga soklite liidese kaudu.

- Protokollid on programmeerija eest varjatud

Kui transpordikihi protokollid muutuvad, ei pea programmi muutma.